-Document classification is fundamental task in NLP, allowing us to categorize text based on their content.

-By analyzing keywords and phrases, classification models can be trained to recognize different patterns, mainly used in customer feedback to judge public opinion and do sentiment analysis.

-It also plays a significant role in identifying spam email, identifying common patterns, key words and links.

-It can also be used to assign subject/categories, authorship identification, age/gender identification and language identification.

-Document classification takes a document as an input, assigns it to a fix set of classes, the output is the predicated class from the set.

-The vector space model is a model where each document is represented as a single vector, this model is mainly used in text mining and information retriever.

-The matrix-based model each document is represented as a matrix N by P where N is the maximum number of words in a document and P is the embedding dimension of each word.

-Bayes rule $P(C_i|x_1, x_2) = \frac{P(x_1,x_2|C_i)\cdot P(C_i)}{P(x_1,x_2)}$ let's us calculate the probability of a class given certain feature (x1,x2)

-Naive lets us assume that all variables are all statistically independent of each other. $=\frac{P(x_1|C_i)\cdot P(x_2|C_i)\cdot P(C_i)}{P(x_1,x_2)}$

-Maximum likehood estimates (MLE) to determine probability's based on frequencies of word in each class.

-k-Nearest neighbor (KNN): method that relies on concept of similarity between data points. It applies the concept that similar things or people tend to group together. Assuming that data points with similar features will also have a similar classification.

-Step 1 of the process of KNN classification is first to prepare the reference data by defining the attributes/features of our data points (using bow representation for ex). We also need to ensure that we have enough records from each class to keep accuracy.

-Step 2: define the similarity measure, similarity is inversely proportional to the distance.

distance$\big(P = (x_1, x_2, \dots, x_n), Q(y_1, y_2, \dots, y_n)\big) = (\sum_{i=1}^{n} | x_i - y_i|^p)^{\frac{1}{p}}$ when p = 2 we get the Euclidian distance (often used in KNN).

-Step 3 we initialize the set of candidate value for k If k is too small, then the classification will be highly locally sensitive (over-fitting). If k is too large, then it will lose the ability to capture the local structure (under-fitting). A proper k should be chosen among a set of candidates. Use the validation data.

-Step 4 : determine the combining rule, majority voting (consider the most common class among the neighbors) vs weighted voting (gives more importance to closure neighbors)

-step 5: find the best k value using validation dataset that minimizes the error rate, this lets us improve our model's accuracy.

-KNN issue 1: Normalization or scaling must be done before finding the k-nearest neighbor

-KNN issue 2 : Cut off, consider the prior probability of each class